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COURSE: Data Science

Exercise 1: SQL Fundamentals: SELECT & Filtering.

1. SELECTING & SORTING.

1. SELECT *

FROM employees

results	ID	first_name	last_name	department	Salary	Hire_date	City
	1	Alice	Green	IT	70000	2020-01-10	Johannesburg
	2	Brian	Lee	HR	45000	2019-03-22	Cape Town
	3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
	4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
	5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department
FROM employees;

results	department
	IT
	HR
	Finance
	Marketing

3. SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

Results:	first_name	last_name
	Eva	Maidoo
	Alice	Green
	Cathy	Zulu
	David	Motoena
	Brian	Lee

4. SELECT id, first_name, last_name, Salary,
 FROM employees
 ORDER BY Salary DESC
 Limit 3;

Results:	id	first_name	last_name	Salary
	5	Eva	Maidoo	72000
	1	Alice	Green	7000
	3	Cathy	Zulu	65000

② Filtering with WHERE

5. SELECT id, first_name, last_name, department
 FROM employees
 WHERE department = 'IT';

Results:	id	first_name	last_name	department
	1	Alice	Green	IT
	5	Eva	Maidoo	IT

6. SELECT id, first_name, last_name, department, Salary
 FROM employees
 WHERE department = 'finance' AND Salary > 60000

Results:	id	first_name	last_name	department	Salary
	3	Cathy	Zulu	finance	65000

7. SELECT id, first_name, last_name, department
FROM employees

WHERE department = 'HR' OR department = 'Marketing';

Results:	id	first_name	last_name	department
	2	Brian	Lee	HR
	4	David	Mokoena	Marketing

8. SELECT id, first_name, last_name, department
FROM employees

WHERE department = 'IT';

Results:	id	first_name	last_name	department
	2	Brian	Lee	HR
	3	Cathy	Zulu	finance
	4	David	Mokoena	Marketing

9. SELECT id, first_name, last_name, department
FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

Results:	id	first_name	last_name	department
	1	Alice	Green	IT
	2	Brian	Lee	HR
	3	Cathy	Zulu	Finance
	5	Eva	Maidoo	IT

10. SELECT id, first-name, last-name, department, Salary, City
FROM employees
WHERE department = 'IT'
AND Salary > 6500
AND city = 'Johannesburg';

Results:

id	first-name	last-name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Maidoo	IT	72000	Johannesburg